

Subcortical Mechanisms in Language: Lexical–Semantic Mechanisms and the Thalamus

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Four previously published cases of dominant thalamic lesion in which the author has participated are reviewed to gain a better understanding of thalamic participation in lexical–semantic functions. Naming deficits in two cases support Nadeau and Crosson's (1997) hypothesis of a selective engagement mechanism involving the frontal lobes, inferior thalamic peduncle, nucleus reticularis, and other thalamic nuclei, possibly the centromedian nucleus. This mechanism selectively engages those cortical areas required to perform a cognitive task, while maintaining other areas in a state of relative disengagement. Deficits in selective engagement disproportionately affect lexical retrieval based on semantic input, as opposed to lexical and sublexical processes, because the former is more dependent upon this attentional system. The concept of selective engagement is also useful in understanding thalamic participation in working memory, as supported by data from one recent functional neuroimaging study. Other processes also may be compromised in more posterior thalamic lesions which damage the pulvinar but not other components of this selective engagement system. A third case with aphasia after a more superior and posterior thalamic lesion also had oral reading errors similar to those in neglect dyslexia. The pattern of deficits suggested a visual processing problem in the early stages of reading. The fourth case had a category-specific naming deficit after posterior thalamic lesion. Taken together, the latter two cases indicate that the nature of language functions in more posterior regions of the dominant thalamus depends upon the cortical connectivity of the thalamic region. Together, findings from the four cases suggest that thalamic nuclei and systems are involved in multiple processes which directly or indirectly support cortical language functions. © 1999 Academic Press

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The lesion method involves examining cognitive functions after a brain region has been damaged. If the region is involved in a specific function such as language, then, the pattern of deficits will give some clues as to the role of the region in the cognitive function. In this respect, deficits caused by vascular lesion of the thalamus can be useful for determining the role of the thalamus in language. This review will focus on four cases of language dysfunction after dominant thalamic lesion. Three of these cases were examined at the University of Florida in the last few years (Crosson, Moberg, Boone, Rothi, & Raymer, 1997; Raymer, Moberg, Crosson, Nadeau, & Rothi, 1997). One case was published in *Brain and Language* in 1986. Findings from these cases offer the opportunity to examine various aspects of lexical and semantic processing in which the thalamus may participate. Because complete descriptions of these cases have been published elsewhere, data are summarized, and this article focuses on integration of the findings and what together they tell us about the role of the dominant thalamus in language. Based upon analysis of these cases, as well as recent theoretical works and other supporting data, it is maintained that specific thalamic structures are critical elements in an attentional ("selective engagement") system that, for language, has its greatest effects on lexical retrieval based on semantic processing. Selective engagement also plays a role in working memory. However, this system does not explain all cases of thalamic aphasia. Other thalamic mechanisms may affect different attentional and/or associational processes that impact lexical and semantic functions.

THALAMIC MECHANISMS INVOLVED IN SELECTIVE ENGAGEMENT

In order to facilitate our discussion, exploration of these four cases is prefaced with two theoretical issues. The first deals with a model of lexical-semantic processing which will guide analysis of our cases; the second issue regards previous models of thalamic functions in language. The model of lexical-semantic processing (Fig. 1) has been explicated in a number of places, including the work of Ellis and Young (1988). The components of the model include four lexicons: a phonologic input lexicon which recognizes familiar spoken word forms, an orthographic input lexicon which recognizes familiar written word forms, a phonologic output lexicon which contains word forms for spoken language, and an orthographic output lexicon which contains word forms for writing. It has become common in many places to think of these lexicons as dealing primarily with word forms and not with meaning. Other sublexical processes are necessary to understand or produce spoken or written words; we shall discuss such components as needed. The model also includes a structural description system which recognizes familiar objects.

It is the semantic system, however, which deals with meaning. The organization of the semantic system is a controversial topic. It is generally assumed

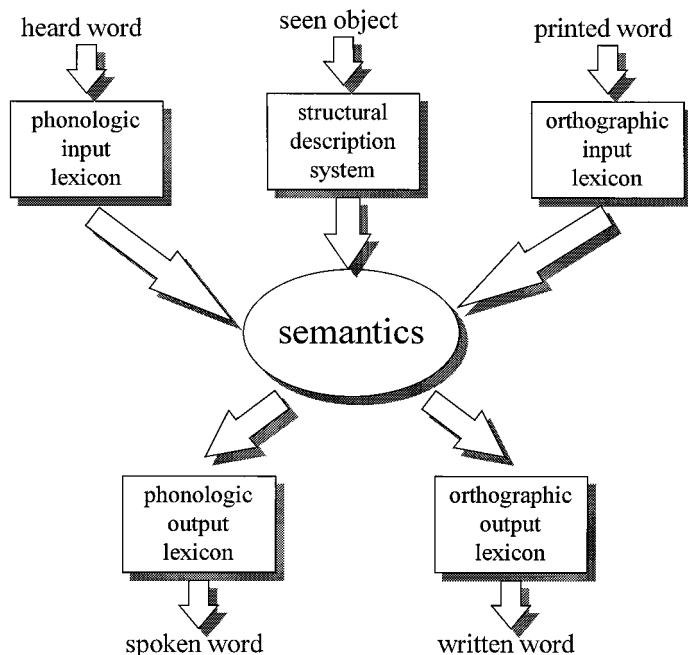


FIG. 1. Schematic representation of interface of semantic system with lexical systems.

to be broadly distributed in the brain and involves information from multiple modalities. Some recent evidence suggests that various taxonomic categories within the semantic system draw differentially on different types of knowledge, such as visual form or function (e.g., Damasio, Damasio, Tranel, & Brandt, 1990; Farah, McMullen, & Meyer, 1991; Hart & Gordon, 1992; Martin, Wiggs, Ungerleider, & Haxby, 1996; McCarthy & Warrington, 1988). At the interface between the input lexical or structural description systems, on the one hand, and the semantic system, on the other hand, meaning is assigned to words or objects. At the interface between the output lexical systems and the semantic system, words are selected to convey a particular meaning. Evidence also suggests that processes involving the interface between the lexical and semantic systems may be compartmentalized by semantic category (e.g., Farah & Wallace, 1992) or by part of speech (e.g., Damasio & Tranel, 1993; Miceli, Silveri, Villa, & Caramazza, 1984; Zingeser & Berndt, 1988). However, others have explained category-specific naming deficits by word frequency, familiarity with a class of objects, and/or complexity of constructs (e.g., Funnell & Sheridan, 1992; Stewart, Parkin, & Hunkin, 1992). I shall apply this model of lexical-semantic functions to thalamic language dysfunction shortly and address the compartmentalization issue later in the article.

Before doing so, we must address the second theoretical issue, a model of how the thalamus participates in language. This model involves a specific kind of attention. The term attention here is used in its broadest sense, to apply to focusing on internal stimuli (i.e., thoughts) as well as external stimuli. Attentional theories for thalamic aphasia are not new. For example, Riklan and Cooper (1975) and McFarling, Rothi, and Heilman (1982) have suggested that higher-level language functions require greater diffuse energizing from thalamic centers. In 1983, Ojemann suggested that "thalamic alerting processes selectively activate the specific discrete cortical areas (mosaics) appropriate to the particular language processing" (p. 203). He had suggested that nonspecific cortical afferents from the intralaminar nuclei might provide the mechanism. In a more recent paper, Nadeau and Crosson (1997) have implicated a system in thalamic aphasia which includes the frontal lobes, the nucleus reticularis, the inferior thalamic peduncle which connects the frontal lobes with the anterior portions of the nucleus reticularis, and possibly the centromedian nucleus or other thalamic nuclei (Fig. 2). (See Nadeau & Crosson [1997] for all the empirical and conceptual references supporting this theory.)

The nucleus reticularis is a thin sheet of GABAergic neurons which surrounds the thalamus and projects to most nuclei of the thalamus, except the anterior nuclei. The inferior thalamic peduncle connects the frontal lobes with the thalamus and nucleus reticularis. The inferior thalamic peduncle passes through the nucleus reticularis at a position inferior and anterior to the ventral anterior nucleus of the thalamus. This part of the nucleus reticularis has a dense neuropil and is rich in synaptic connectivity. It may connect to more posterior thalamic nuclei such as the centromedian nucleus directly (Jones, 1985), and Nadeau and Crosson (1997) have speculated that more posterior portions of the nucleus reticularis, which project to posterior thalamic nuclei, may make contact with collaterals from the inferior thalamic peduncle through long dendritic bundles (Scheibel & Scheibel, 1972) that may extend to the anterior region of the nucleus reticularis. The centromedian nucleus, along with other thalamic intralaminar nuclei, projects diffusely to the cerebral cortex, and thus could affect a number of cortical areas. According to Nadeau and Crosson (1997), this fronto-inferior thalamic peduncle-nucleus reticularis-centromedian system works to selectively engage those cortical units necessary to perform various tasks while maintaining unnecessary cortical units in a relatively disengaged state. There is an alternative to the suggestion of Nadeau and Crosson that the centromedian nucleus is involved in selective engagement. This alternative is that the nucleus reticularis accomplishes selective engagement through its influence on other thalamic nuclei, such as the pulvinar, which are richly connected to language cortex.

Selective engagement serves to increase efficiency within the activated cortical unit and enhance the communication between various units that are

FRONTAL CORTEX

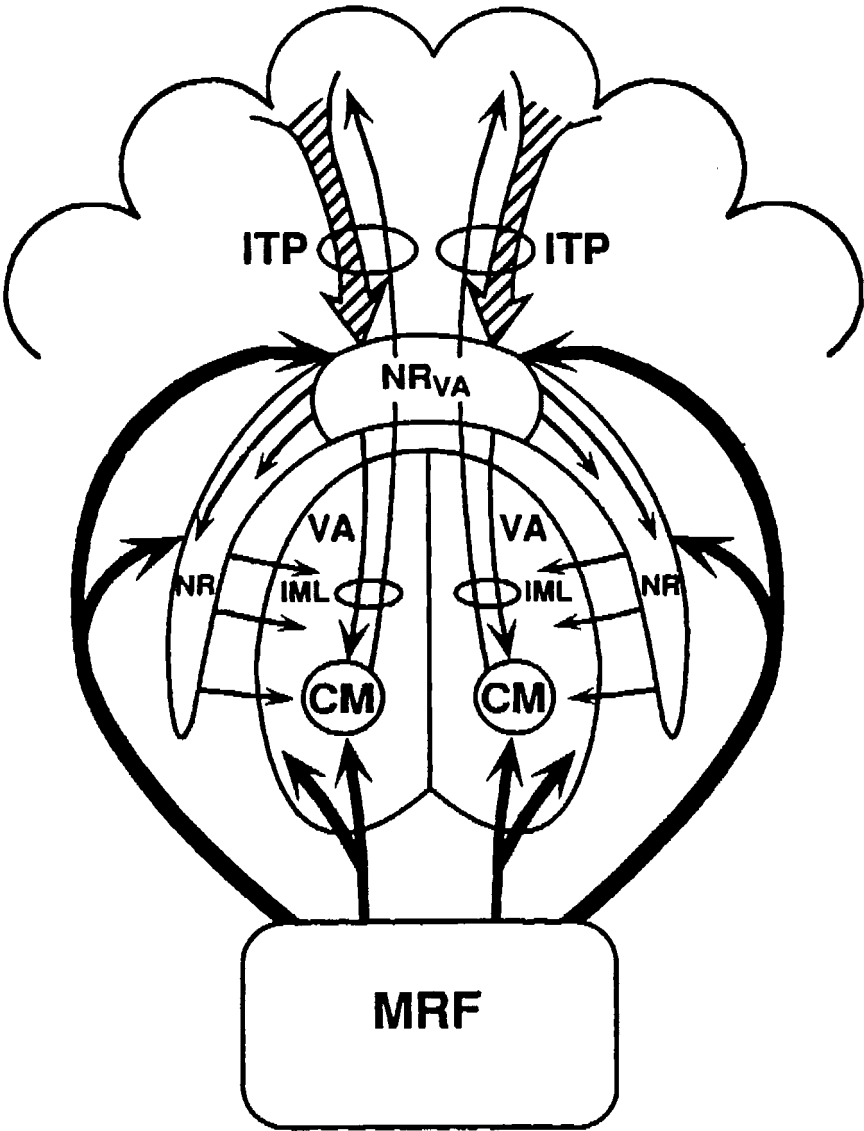


FIG. 2. Schematic representation of anatomic substrate for selective engagement. ITP, inferior thalamic peduncle; NR_{VA}, nucleus reticularis ventral anterior; NR, nucleus reticularis; VA, ventral anterior nucleus of thalamus; IML, internal medullary lamina; CM, centromedian nucleus of thalamus; MRF, midbrain reticular formation. (From Nadeau & Crosson, 1997.)

engaged. The result will be to sharpen the difference in activation between representations during cognitive processing. The less automatic a particular cognitive activity (i.e., the more attention it requires) the more vulnerable the activity will be to deficits in selective engagement. This model explains how the frontal lobes might direct attention via thalamic nuclei.

I would now like to explore two cases in which the hypothesis of selective engagement is relevant (Raymer et al., 1997). Case BD was a 45-year-old, right-handed woman with a high-school education who worked as a book-keeper, and Case WT was a 59-year-old, right-handed man with a college education who was a retired salesman. BD had an infarction in the left tuberothalamic artery distribution which included the ventral anterior portion of the nucleus reticularis and inferior thalamic peduncle (Fig. 3a). WT had an infarction in the left paramedian artery distribution which missed the ventral anterior nucleus reticularis and inferior thalamic peduncle but hit the centromedian nucleus (Fig. 3b). Thus, the lesions in these two patients involved different portions of the frontal-inferior thalamic peduncle-nucleus reticularis-centromedian system.

The inference that their language deficits derived from lesions to the same system, albeit in different places, is bolstered by the fact that they showed nearly identical naming deficits when tested 4 (WT) to 5 (BD) months post stroke. Both patients were administered a lexical-semantic battery, the Florida Semantics Battery (Rothi, Raymer, Maher, Greenwald, & Morris, 1991), designed to locate the part of the lexical-semantic system which is impaired during naming deficits. The battery includes seven tasks of lexical processing: oral picture naming, written picture naming, naming to auditory definition, oral word reading, writing to dictation, auditory word/picture matching, and written word/picture matching. The battery was designed to test the model of lexical-semantic functions described earlier in this article. If a deficit is semantic, patients should show poor performance on all picture naming and comprehension tasks because they cannot process meaning. Repetition and writing to dictation may be intact because they can be done on the basis of lexical knowledge alone. If the deficit was primarily lexical, then individual lexicons might be differentially affected. For example, a deficit in the phonological output lexicon would impact oral picture naming and naming to auditory definition because these activities require access to spoken word forms. Oral word reading might be performed through sublexical processes, at least for orthographically regular words, and other tasks from this battery would not require access to spoken word forms.

Actually, the pattern of performance for BD and WT suggested neither a purely semantic nor a purely lexical deficit. Both patients had difficulties relative to controls primarily in the three naming tasks: oral picture naming, written picture naming, and oral naming to auditory definition (Fig. 4). Thus, in tasks which require retrieval of items from the phonological output lexicon based on semantic information, the patients showed impairment. On tasks

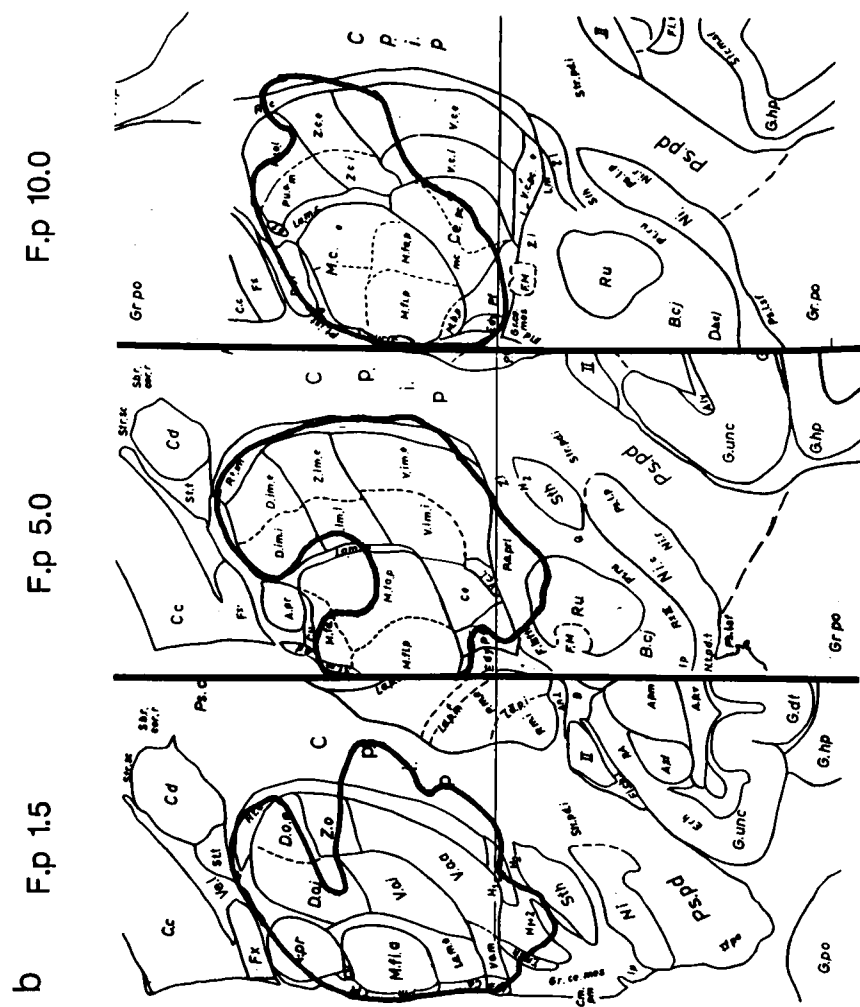
which require that phonological or orthographic output forms be retrieved based on the orthographic or phonologic input forms, respectively, performance was normal, indicating intact lexical processing. Auditory word/picture matching and written word/picture matching were also normal. The ability to match auditory and written forms to correct pictures implies not only intact phonologic and orthographic input lexicons but also an intact semantic system.

Further analyses were helpful in determining the nature of the problem. Low frequency words were generally more difficult to name for both patients. Naming errors could be divided into items which shared visual and semantic properties with the target (e.g., fork for spoon), errors which shared semantic (but not visual) properties with the target (e.g., guitar for trumpet), omissions, and other errors. The majority of errors for both cases (BD = 84%; WT = 63%) were either visual/semantic or semantic. There were no purely visual errors for either patient. The presence of purely semantic errors and absence of visual errors suggests the misnaming is due primarily to processes which somehow involve the semantic system.

We hypothesized that in both cases naming deficits were due to a failure of the frontal-inferior thalamic peduncle-nucleus reticularis-centromedian system to selectively engage cortical mechanisms necessary for lexical retrieval based upon semantic information. This failure of selective engagement will cause decreased efficiency of the lexical-semantic interface necessary for naming. The frequency effect most likely was caused because low frequency items are defined by fewer associations; i.e., they are less redundantly represented, and therefore are more vulnerable to disruption. The predominance of semantic errors indicates that degraded information somewhere at the interface of the semantic system and the output lexicons leads to a failure in selecting the precise lexical item to describe the semantic concept.

If this interpretation is correct, then why does selective enagement primarily affect lexical output only when semantic processes are involved? Should we not expect a selective engagement failure to affect lexical output even

FIG. 3. (a) BD's thalamic lesion. Transverse sections of left thalamus showing BD's infarction in the tuberothalamic artery territory. The lesion included the ventral anterior (lateral polar) nucleus (Lpo) and the ventral lateral nucleus just posterior to the ventral anterior nucleus. The nucleus reticularis was also damaged. It is the thin band lateral and anterior to the thalamic nuclei and medial to the internal capsule (Cp.i.p and Cp.i.g). The inferior extension of the lesion (right-most panel) suggests that the more ventral portion of the anterior nucleus reticularis was also involved. (b) WT's thalamic lesion. Coronal sections of left thalamus showing WT's infarction in the paramedian artery territory. Among other nuclei, the lesion included the centromedian nucleus (Ce). All panels also show some involvement of the nucleus reticularis, the thin band between the other thalamic nuclei and the internal capsule (Cp.i.p). (From Nadeau & Crosson, 1997.)



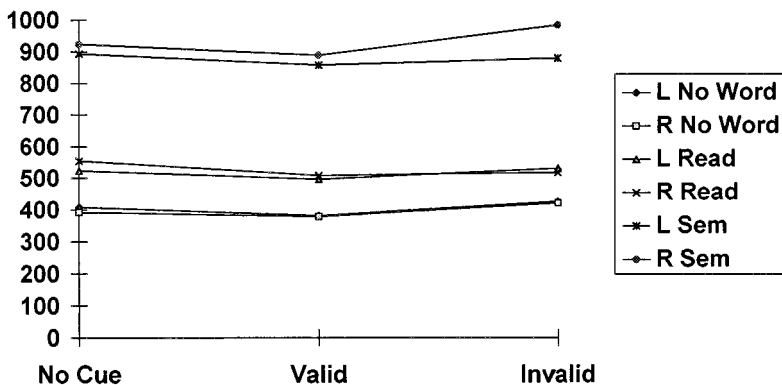


FIG. 4. Lexical vs. semantic interference. Generation of semantic associate interferes with processing of visual information in both visual fields. Petry (1995) required normal subjects to read a word or generate a semantic associate for it prior to the presentation of cues in a visual attention task. The condition shown here was for a 100-ms interval between presentation of the word and presentation of the visual cue and a 100-ms interval between presentation of the cue and onset of the target. The 250-ms word-to-cue interval generated similar results, indicating greater interference of the semantic associate condition on visual attention.

when primarily lexical processes are involved? I would like to suggest that lexical output based upon semantic processing is more dependent upon attentional processes than more purely lexical tasks.

What is the evidence for this position? Among other data, Margaret Petry's (1995) doctoral dissertation, done in our laboratory, suggests this is exactly the case. Petry adapted a visual attention task developed by Posner and his colleagues to look at the interface of attentional processes on the one hand and lexical or lexical-semantic functions on the other. The original Posner task required subjects to respond as quickly as possible by pressing a response key when a target consisting of an asterisk appeared on a computer monitor. Subjects were required to focus on a central fixation point consisting of a plus. To the right and left of the plus were two boxes. On most trials, a cue is given when one of the boxes brightens. After most cues, the target appears in the box that has been brightened. These are called valid trials. On some cued trials, however, the target appears in the uncued box. These are called invalid trials. The interval between the cue and the target is either 100 or 800 ms. On a few trials the target appears when no cue has been given. These are called uncued trials. Normally, valid trials yield the quickest reaction time, invalid trials yield the slowest reaction time, and uncued trials are intermediate. Patients with parietal lesions show abnormally slow responses to invalid trials in which the target is in the visual field contralateral to their lesion.

Petry added two conditions to this paradigm. Either 100 or 250 ms before the visual cue, a word replaced the central fixation point. Under one condi-

tion, subjects simply had to read the word aloud, but under the other condition, subjects had to give a semantic associate for the word. The reading could be done on a lexical basis (a large proportion of the words were orthographically irregular), but giving a semantic associate required semantic processing. We had expected to find some differences for responses to targets in the right visual field given the linguistic nature of the interference. Petry's analyses have indicated, however, the most striking effect was that all semantic trials produced an extreme elevation of reaction time over other trials, for both visual fields and any type of cue (e.g., see Fig. 4). Reading trials were also elevated over normal trials, but not nearly as much as semantic trials. The implication is that forcing the subjects to retrieve a lexical item on the basis of its semantic characteristics is significantly more dependent upon attentional processes than the reading task, probably accomplished primarily through lexical processing. This finding helps to explain why our two patients with tuberothalamic and paramedian lesions, respectively, had deficits only on tasks requiring them to retrieve lexical items on the basis of semantic information. Since this type of lexical-semantic processing is more dependent upon attentional mechanisms, it is more vulnerable when selective engagement mechanisms are interrupted.

It should be noted that these two patients were tested after the acute phase of recovery, and despite their naming deficits, their language was adequate for functional communication. Deficits were much more severe in the phase of recovery immediately post ictus, and some impairments in functions besides word finding were noted at that time, as is common in thalamic aphasia. The current explanation of thalamic functions affords an explanation for these other, less severe deficits in the more acute phases. Effects on the selective engagement system are most severe at this point in time and, therefore, affect functions other than word retrieval at the lexical-semantic interface.

Regarding the effects we documented in later phases of recovery (4 to 5 months post onset), the question arises: How does selective engagement enhance cognitive processes responsible for lexical selection at the lexical-semantic interface? Whether selective engagement might specifically affect any semantic feedback mechanism such as that proposed by Crosson (1985, 1992) is also an interesting question. To explore this possibility, we might turn to cognitive science to ascertain if any models address the issue of feedback during lexical selection at the lexical semantic interface. In this regard, the assumptions of parallel distributed processing (PDP) are (e.g., Rumelhart & McClelland, 1986) are worth exploring. In PDP models, both feedforward and feedback mechanisms between different levels of processing are assumed to exist.

For the sake of simplicity, let us assume the simplest interface between the semantic system and the output lexicon (Fig. 5). Under such a model, an item in the semantic system would be defined by all its characteristics. For example, animate, four legs, tail, furry, carnivore, stealthy, lands on feet,

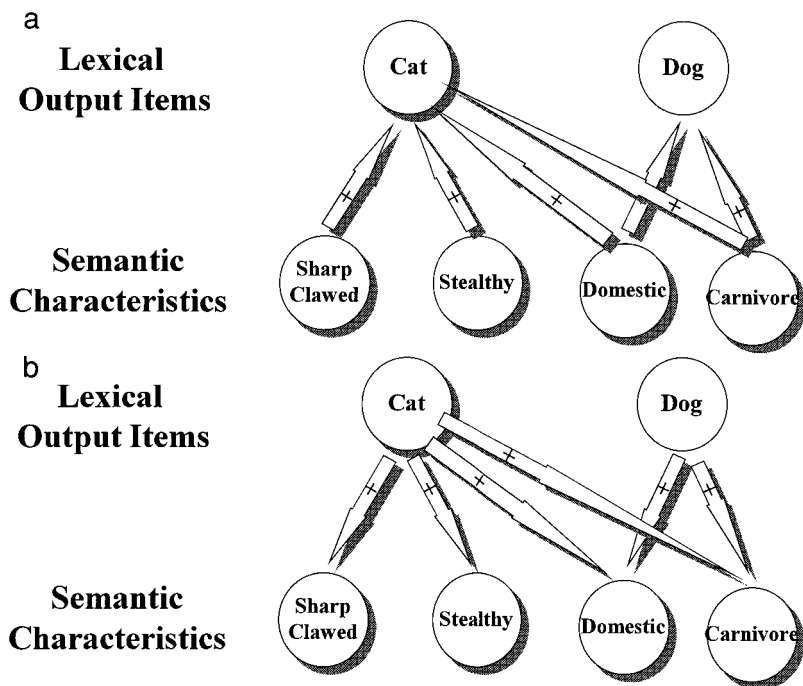


FIG. 5. Simplest model of interface between semantic system and output lexicon. (a) Semantic characteristics feedforward to activate cat more than dog. (b) Once lexical items are activated, they in turn, feedback to increase activation of semantic constructs defining cat and dog.

purrs, pointed ears, meows, etc., will define cat. These semantic characteristics are capable not only of evoking the word "cat" in the output lexicon, they are also capable of evoking the image of a cat in the visual system. Further, not all of the characteristics need be invoked to implicate cat. In most instances, a few of the characteristics, and in some cases even one highly specific characteristic (e.g., meows, purrs) will be adequate to evoke the word "cat" or the image of a cat. In the case of visual naming, the picture of a cat is recognized, its semantic characteristics are activated, and the correct word is retrieved from the output lexicon.

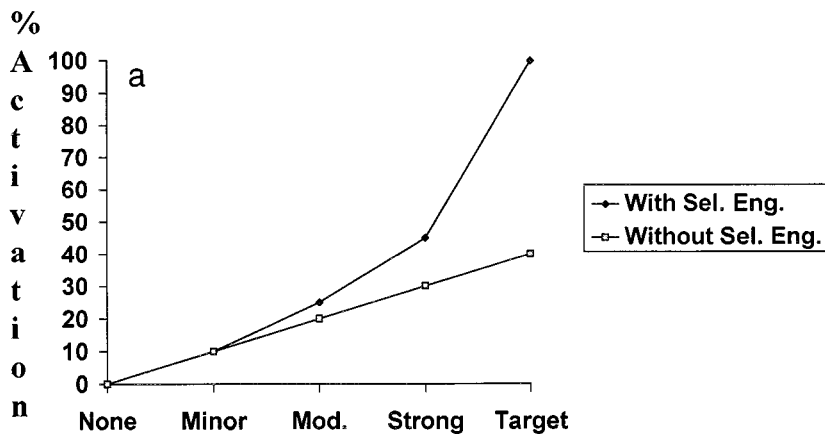
If one accepts assumptions from PDP and similar frameworks, the process of selecting an item from the output lexicon involves not only activation of the correct item in the output lexicon, but also activation of items that are closely related from a semantic standpoint. For example, if given the characteristics "carnivore," "four legs," "stealthy," and "purrs," feedforward processes from all these characteristics will tend to activate "cat." However, "dog" also may be activated by "carnivore," and "four legs" (Fig. 5a). Inhibition is also assumed to occur such that "dog" may be weakly inhibited

by “stealthy” and strongly inhibited by “purrs.” Feedback processes from the lexical items to the semantic level will also tend to activate semantic characteristics (Fig. 5b). When a defining characteristic such as “purrs” is included, the feedback processes may not be so important. But under conditions of partial information when the characteristics are more ambiguous, feedback processes may be more important in reaching a reasonable selection from the output lexicon. For example, “carnivore,” “four legs,” and “stealthy” activate the lexical item “cat” only slightly more than “dog,” and not enough to activate “cat” above the threshold needed to produce an output at the lexical level. However, the slight advantage given to “cat” will tend to activate “purrs” more than “dog” activates “barks.” Further, within the semantic level, “stealthy” will tend to weakly inhibit “barks” while strengthening “purrs,” giving “purrs” a further advantage over “barks.” This advantage for “purrs” now feeds forward to strengthen the activation of “cat” while inhibiting “dog.” Eventually, “cat” is selected as the appropriate lexical item because its activation level is sufficiently above that of competing responses.

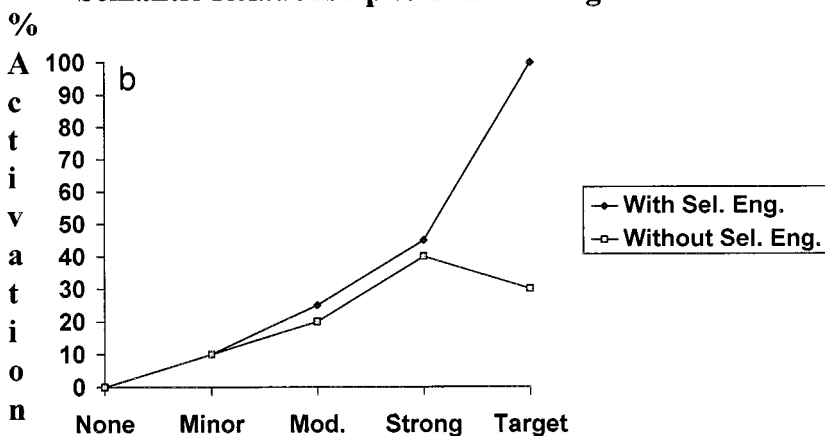
The function of normal selective engagement may be to enhance differences between the activation level of an accurate response versus that of competing, closely related but inaccurate, responses. Under such conditions with normal selective engagement, the probability of selecting an inaccurate competing response is low (Fig. 6a). When selective engagement mechanisms are impaired, both feedforward (semantic to lexical) and feedback (lexical to semantic) processing at the lexical–semantic interface would be decreased in efficiency. This would result in an increased probability of selecting inappropriate competing lexical items because they are not adequately distinguished in activation level from an accurate response (Fig. 6b).

In addition to facilitating the selection of a precise lexical item to express a semantic concept, Nadeau and Crosson (1997) suggested that selective engagement mechanisms can be used to hold lexical–semantic information on line in the service of working memory. In a recent functional magnetic resonance imaging (fMRI) study performed at the Medical College of Wisconsin (Crosson et al., 1999), findings indicated not only that anterior portions of the thalamus are involved in working memory for lexical and semantic information, but also that the participation of the thalamus is unique to circumstances when subjects relied on use of a complete lexical item as opposed to sublexical processing.

In this study, subjects were asked to hold three words in working memory and to indicate if each word of a series matched any of the memory words based on semantic, phonological, or orthographic characteristics. The semantic task involved determining if a word belonged to one of the categories represented by the memory words; the phonological task involved determining if a word rhymed with any memory word; and the orthographic task involved determining if a word had the same last three letters as any memory



Semantic Relationship to Lexical Target



Semantic Relationship to Lexical Target

FIG. 6. Activation of lexical items with and without selective engagement. Ordinate indicates percentage of optimal activation; abscissa indicates the degree of semantic relationship activated items have with the target. The stronger the semantic relationship of a lexical item with the target, the higher its activation level. (a) Selective engagement serves to amplify the activation level of the target. Without selective engagement, the target lexical item is often chosen even though the difference in activation between it and a strong semantic associate is reduced. (b) However, without amplification of the target item activation by selective engagement, strong semantic associates may occasionally show greater activation than the target, leading to selection of the semantic associate. The result is a semantic paraphasia.

word. Structured postexperimental interviews indicated that subjects processed whole lexical items on the semantic and phonological tasks, but on the orthographic task, they used sublexical strategies that involved remembering only the last three letters or the sounds they made. When the lexical units were held in working memory on both the semantic and the phonological tasks, there was a significant activation in the anterior thalamus which appeared to extend into the area of the ventral anterior nucleus reticularis. However, when sublexical units were held in working memory, there was no activation of the anterior thalamus. This finding has led us to speculate that selective engagement mechanisms involving the thalamus are used primarily to activate preexisting cortical representations, in this case for lexical items. It is of some interest that posterior thalamic mechanisms were activated only on the semantic task, suggesting the posterior thalamus may play a role in language unique to semantic processing. We will discuss other potential roles in language for the posterior thalamus shortly.

Before doing so, however, we should briefly note that there is electrocortical evidence that is supportive of the role of the thalamus in selective engagement. Ojemann and colleagues (Fried, Ojemann, & Fetz, 1981; Ojemann, Fried, & Lettich, 1989) have noted two types of potentials during silent naming of objects that are relevant to our discussion. First, in a silent naming task, there is a slow negative potential in the posterior left frontal lobe, beginning at onset of the picture to be named and reaching maximum about 750 ms later. This shift is maximum in premotor cortex but occurs at posterior prefrontal and motor sites as well (Fried et al., 1981). The second phenomenon is desynchronization in posterior left perisylvian cortex that is most prominent 700 to 1200 ms after onset of the stimulus to be named (Fried et al., 1981; Ojemann et al., 1989). Ojemann attributed this desynchronization to the nonspecific thalamocortical activating system. The temporal relationship of the different events in frontal and posterior perisylvian cortex of the dominant hemisphere, as well as Ojemann's interpretation of them, is entirely consistent with Nadeau and Crosson's (1997) model of selective engagement, in which the frontal cortex engages the specific cortical mechanisms needed for lexical-semantic functions through thalamic mechanisms.

OTHER THALAMIC MECHANISMS RELATED TO LANGUAGE

If the story ended here, we would have a satisfactory unitary explanation for thalamic aphasia, but consistent with most brain functions, the plot is somewhat more tortuous. The question that arises is whether dominant thalamic nuclei have any functions in language other than selective engagement. We have seen two cases that suggest this may be the case. In 1986, we reported a case of an 82-year-old, right-handed male, RD, with an 8th-grade education. He had a hemorrhagic infarction of the left lateral nucleus extending into the superior left pulvinar which caused a significant aphasia

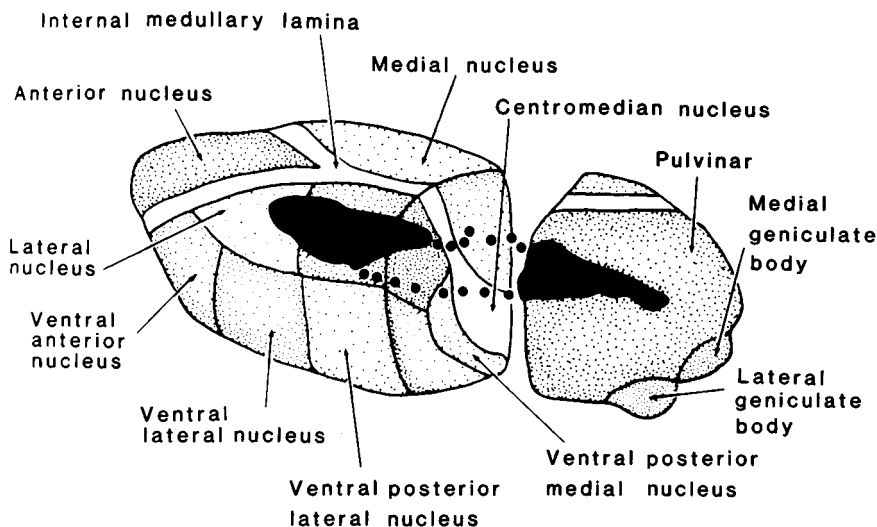


FIG. 7. RD's thalamic lesion. Three-dimensional model of thalamus showing location of RD's hemorrhagic infarction in the lateral nucleus and the superior pulvinar. (From Crosson et al., 1986.)

(Fig. 7). Findings from the Boston Diagnostic Aphasia Examination revealed a fairly typical thalamic aphasia. The patient's speech and language were generally fluent; semantic paraphasia sometimes deteriorated into jargon; repetition was minimally impaired; and comprehension was less impaired than output. Since the patient's lesion falls outside the fronto-inferior thalamic peduncle-nucleus reticularis-centromedian system, we can conclude that lesions in the thalamus outside that system may cause aphasia. On the surface, RD's aphasia was similar to the acute symptoms of BD, who was described in the previous section of this review. This similarity raises the issue of whether there is a relationship between mechanisms in the anterior portions of the thalamus (i.e., nucleus reticularis) discussed in the previous section and mechanisms involving the pulvinar. We have already discussed the possibility that the nucleus reticularis actually influences the pulvinar as opposed to the centromedian nucleus as far as language functions are concerned.

However, this patient had some problems with reading that implicate spatial mechanisms in reading as well as lexical-semantic functions. In 1986, we reported that during spoken language, the patient most often produced semantic paraphasias, like most cases of thalamic aphasia. However, when reading aloud, the patient produced real words which had what we originally interpreted as a phonologic similarity to the target word. We were able to reanalyze his responses to the Boston Diagnostic Aphasia Examination. Although the number of items is not optimal, we can get some idea of the

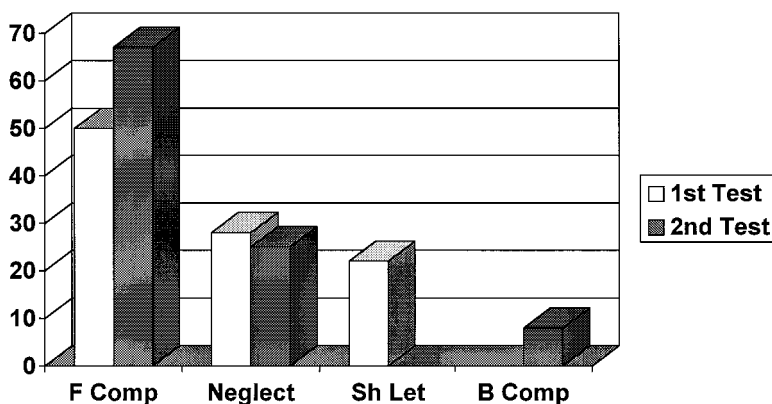
(1st Test: $n = 18$; 2nd Test: $n = 12$)

FIG. 8. RD's visual/phonological reading errors. RD's reading errors on Boston Diagnostic Aphasia Examination subtests usually bore some visual and phonological similarity to the target. The graph indicates the percentage of these errors that were forward completion errors (F Comp), neglect errors, errors sharing letters with the target in some other way (Sh Let), and backward completion errors (B Comp) on the first assessment (1st Test) and the second assessment (2nd Test).

nature of his reading problem. Most reading errors (58% on initial testing, 67% on follow-up testing) had some visual and phonologic similarity to targets. Most of these errors were similar to what Hillis and Caramazza (1990) termed forward completion errors (Fig. 8): They had at least the first two left-hand letters the same as the target, had one or more letters on the right of the response different from the target, and were real words. This type of error indicates that the patient was aware that the word contained more letters to the right of the initial few, but he was unable to get them correct. The question arises whether he inadequately processed the visual stimulus or he had an inadequate internal representation of the stimulus. According to Hillis and Caramazza (1990), if the internal representation is inadequate, the patient will make similar errors for words that are orally spelled, whereas if the error is in processing the visual stimulus and the internal representation is undisturbed, the patient will do well in retrieving words that are orally spelled for him. RD's performance improved dramatically during oral spelling, suggesting that his errors were not in the internal representation of the word but in the earlier visual processing of the stimulus. We conclude that the **thalamic lesion affected perceptual processes in the early stages of reading in addition to semantic processes in spoken language output.**

A few comparisons to other cases should be made. In Hillis and Caramazza's (1990) cases of neglect dyslexia, their patients did equally poorly on orally spelled words and reading. But, Ellis, Flude, and Young (1987) re-

ported a case whose responses to orally spelled words improved over reading. It should be noted that most cases of neglect dyslexia involve the left side of words after right-hemisphere lesions; however, one of Hillis and Caramazza's (1990) cases had difficulty with the right-hand side of words after a left-hemisphere lesion. Regarding RD's aphasia, we are uncertain if the semantic deficit was of a similar nature to the two cases which were discussed above. The lesion appeared superior to the location of the centromedian nucleus and, therefore, does not appear to have involved the frontal-inferior thalamic peduncle-nucleus reticularis-centromedian system. The dominant pulvinar has connections to the perisylvian language cortex as well as to temporal and parietal cortices involving visual processing (Jones, 1985). However, as just noted, the nucleus reticularis projects to the pulvinar and may modulate its influence on perisylvian neocortex. This may account for similarities in pattern of language deficit to BD in her acute phases. The presence of a visual processing deficit affecting reading suggests that the connections between the pulvinar and visual association cortex in the left hemisphere are important to RD's reading performance.

The fourth case study will raise further questions about the role of the pulvinar in language. Before discussing this case, however, we need to briefly examine the concept of category-specific naming deficit, also referred to as semantically bounded anomia. These terms describe a naming deficit, or sometimes the preservation of naming, for items from a specific semantic category. In some cases, the category is broad such as "living things" and in others it is more specific such as "animals" or "fruits and vegetables." The explanations for this type of deficit have varied. Funnell and Sheridan (1992) provided evidence that an apparent naming deficit for living things in a head injury patient was actually attributable to differences in familiarity with the items to be named. Stewart et al. (1992) similarly demonstrated that an apparent deficit in naming animals in a herpes simplex patient could be eliminated when frequency of the target word, familiarity of the concept, and visual complexity were controlled. Both of these sets of authors concluded that the semantic system was not compartmentalized by taxonomic category. On the other hand, Sacchett and Humphreys (1992) attributed a deficit in naming artifactual objects and body parts to an inability to access semantic information about these items from vision. In addition to the naming deficit, their patient made errors in picture-word matching when semantic distractors were used, and he had difficulty matching items with real-world contextual knowledge. It is also of note that the latter patient's naming was significantly facilitated by phonemic cues, suggesting that the lexical items were available but could not be accessed because of degraded semantic information.

A case of Farah and Wallace (1992) is of interest because of its differences with Sacchett and Humphreys' (1992) case. The former case had specific difficulty naming fruits and vegetables, from pictures, real items, or defini-

tions. However, unlike Sacchett and Humphreys' case, this gentleman did not appear to have a semantic deficit. For example, when given the name of a fruit or a vegetable, he could describe it quite well even when he had failed to name the item correctly. In all cases in which the subject was given a phonemic cue when he failed to name a fruit or a vegetable, he was able to then access the correct name, indicating that the representations from the phonologic output lexicon were available. These authors concluded that their patient's deficit was neither lexical nor semantic. The authors proposed something like hidden units between the semantic system and the phonologic output lexicon that represent clusters of semantically similar words and facilitate lexical retrieval.

Now, we will turn to the case of BC (Crosson et al., 1997). It should be noted that he was tested in 1990, before many of the critical articles on category-specific naming deficits came out; therefore, our data in some areas are comparatively sketchy. BC was a 46-year-old, right-handed man with 14 years of education. He had a hemorrhagic lesion of the left thalamus which affected portions of the pulvinar, posterior limb of the internal capsule, centromedian nucleus, and other nuclei (Fig. 9). Although initially aphasic, he showed very few errors on the Western Aphasia Battery at 59 days post onset. Nonetheless, the patient complained that he had difficulty naming terms related to his medical condition or items around the hospital. When these reports were confirmed by hospital staff, we undertook formal testing of the described deficit.

We constructed five lists to do so: living things; nonliving human-made items; nonliving natural items; human-made medical items; and medical conditions, signs, personnel, and anatomical terms. Because many items could not be easily illustrated visually, a naming-to-definition format was used. Each list had 37 items, and the lists were matched for overall word frequency (Frances & Kucera, 1982). Although there were some differences between medical vs. nonmedical lists with respect to imageability and complexity, analyses indicated that these variables did not account for BC's performance. The lists were also administered to 11 normal controls and 2 anomic patients. It can be seen (Fig. 10) that BC had approximately a 30% error rate in the two medical items categories and close to 100% accuracy for other items. Normal controls had relatively high accuracy on all lists, and anomic patients performed at a level similar to BC's medical items performance on all lists. BC was relatively consistent in those medical items which he missed. Of 22 medical items missed on the first testing 16 were missed on a second testing, and only 2 items not missed on the first testing were missed on the second administration. Thus, BC appears to have a category-specific naming deficit for medical items.

The question regarding why BC has such a deficit with a thalamic lesion is not easily answered by our data. Unfortunately, the information we have about BC's semantic knowledge of the missed items is limited: during our

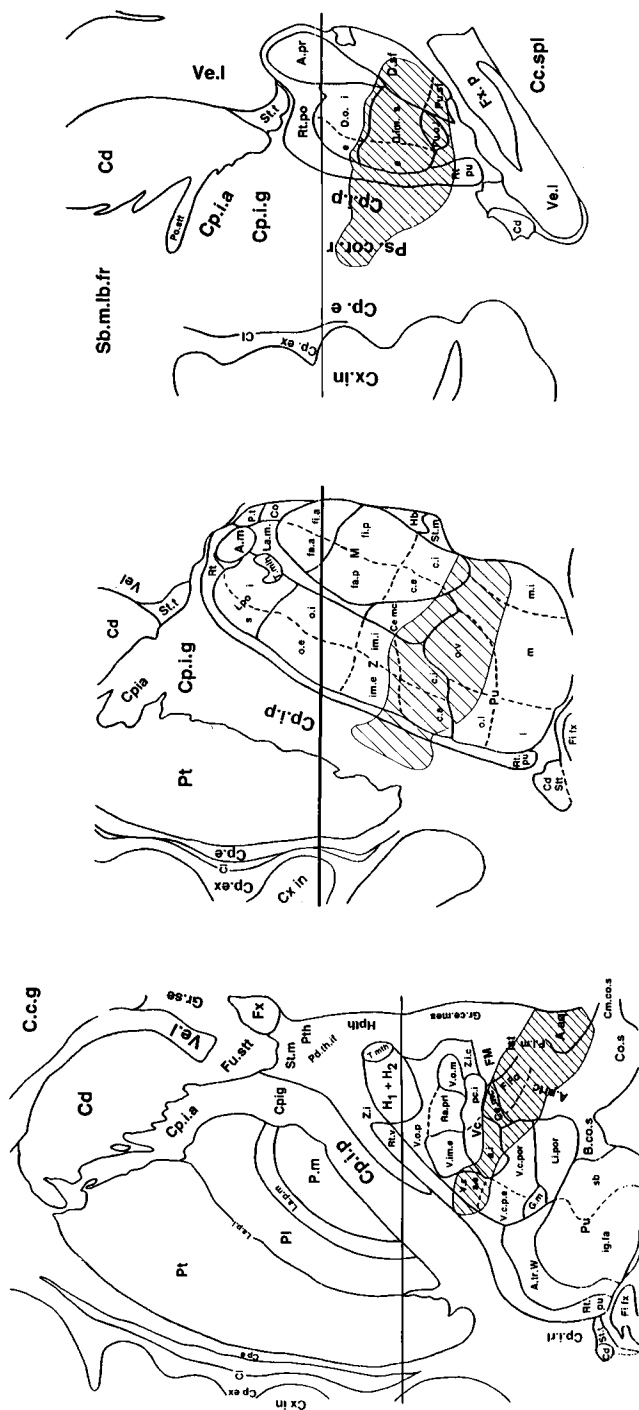


FIG. 9. BC's thalamic lesion. Transverse sections of thalamus showing BC's thalamic hemorrhage which occupies a portion of the pulvinar (Pu) and extends into the internal capsule (Cp.i.p). (From Crosson et al., 1997.)

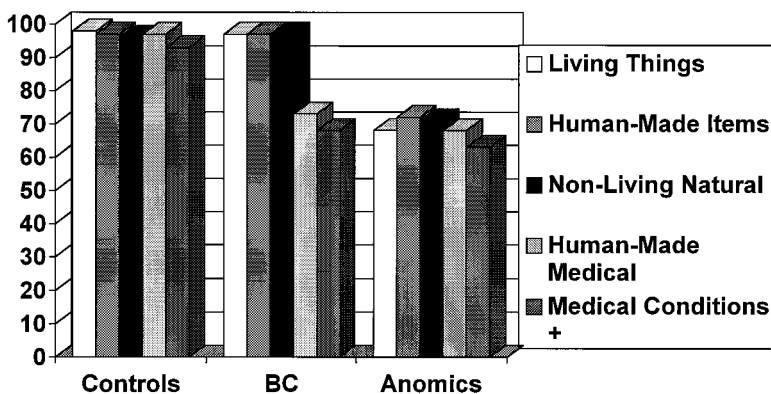


FIG. 10. BC's performance on medical and nonmedical items. BC's performance on medical and nonmedical items is compared to the average performance of 11 controls and the average performance of 2 anomics on the same naming-to-definition lists. BC's error rate was significant only for medical items. (From Crosson et al., 1997).

naming to definition task, he noted that he recognized the definition but just could not name the item. Seventy-seven percent of his errors were failures to produce any response, and 23% of the errors were semantically related items. This pattern is different from that of normals as well as the previous three cases of thalamic lesion in that all tended to produce semantically related errors; however, BC's response pattern was similar to that of the two anommic patients for this specific set of items because anomics showed a similar incidence of failures to respond. Unlike the case of Farah and Wallace (1992), BC did not benefit from phonemic cues; he produced the correct lexical item on only 3 of 22 trials after phonemic cuing. We cannot be certain on the basis of the data we have whether the deficit existed at the lexical or the semantic level of processing, or at the interface between these levels. Since we do not believe lexical or semantic information to be represented per se at the level of the pulvinar or other thalamic nuclei, it appears that thalamo-cortical connections influence processes leading to lexical access.

Why was the anomia semantically bounded in this case of thalamic lesion? It should be noted that two cases of category-specific naming deficit (proper name anomia) after lesion in the dominant tuberothalamic artery territory have been reported (Lucchelli & DeRenzi, 1992; Moreaud, Pellat, Charnallet, Carbonnel, & Brennen, 1995). Lucchelli and DeRenzi suggested that in their case, proper nouns were harder to name because the name of a person or place is often unique to the individual person or place, and other than the unique attributes of that person or place has few links to semantic information. This explanation has little utility for medical items; however, the fact that proper noun anomia occurs with a more anterior thalamic lesion, while

our case involved a more posterior thalamic lesion, indicates that it is not a matter of the connectivity of a single nucleus that determines these category specific deficits. More than likely, it is a matter of what brain systems are invoked in naming the different types of items and how specific thalamic sites interact with the systems. For example, Martin and colleagues (Martin et al., 1996) have shown that systems involved in tool naming overlap, but also diverge with, systems involved in animal naming. The fact that premotor cortex was activated in tool naming more than likely reflects the facts that tools are involved in actions and that the premotor cortex has stored information about actions.

SUMMARY

To briefly recap: What do our four cases tell us about lexical–semantic functions and the thalamus? The following conclusions and questions can be derived:

The data are generally supportive of the notion that the thalamus is involved in selective engagement of cortical mechanisms necessary to perform language tasks. The interface between the semantic system and the output lexicons is most vulnerable because it is more dependent on selective engagement. More purely lexical processes are more automatic and, therefore, less dependent on selective engagement, although some errors may be found at the lexical or sublexical level as well in patients with lesions of the dominant thalamus, particularly in the more acute phase of recovery. Selective engagement probably is accomplished via a system including the left frontal lobe, inferior thalamic peduncle, and nucleus reticularis. Assumptions based on PDP modeling can be used to explain the errors seen in two patients with lesions of this selective engagement system. Ultimately, however, such models must be consistent with what we know about the neurophysiological substrates for cognitive processing. The similarity of BD, with an anterior (polar artery) lesion, and RD, with a posterior thalamic lesion, in their acute stages suggests that the portion of the nucleus reticularis which was affected for BD may actually regulate the pulvinar instead of or in addition to the centromedian nucleus. This part of the model deserves further attention.

The concept of selective engagement can be applied to working memory. Continuing engagement of cortical mechanisms involved in processing specific lexical–semantic information must be extended over time to maintain the information in working memory. In a recent functional imaging study (Crosson et al., 1999), we found the anterior thalamus activated when working memory relied on whole lexical units but not when subjects relied on sublexical units. Thus, the role of the thalamus in selective engagement and working memory may be specific to circumstances when preexisting representations (e.g., lexical items) are invoked and/or maintained in working memory.

The dominant thalamus may play other supportive roles for language functions. Findings from RD indicate that the ability to correctly perceive letters on the right side of the words may be influenced by mechanisms in the dominant thalamus. We would presume the nondominant thalamus might affect perception of letters on the left side of words, since this finding of neglect dyslexia is more common than trouble perceiving the right side of words.

The existence of category-specific **naming deficits with thalamic lesion** raises further questions regarding the role of the thalamus in language. It is highly unlikely that specific categories of lexical items are stored at the thalamic level. The existence of proper name anomia in anterior thalamic lesions (Lucchelli & DeRenzi, 1992; Moreaud et al., 1995) and medical item anomia with posterior thalamic lesion (Crosson et al., 1997) indicates these deficits are not unique to a specific thalamic region. Instead, it would appear that the thalamus is involved in lexical-semantic systems that are needed to name various types of items. Certainly, the cortical connectivity of thalamic nuclei and the role of specific cortical regions in naming items from a taxonomic category will determine the nature of category-specific deficits after thalamic lesion. Functional imaging studies are beginning to shed some light on contributions of specific cortical regions to naming various categories of items. Perhaps this work will eventually contribute to our knowledge about the role of the thalamus in category-specific naming deficits and, by extension, our broader knowledge about the role of the thalamus in language.

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